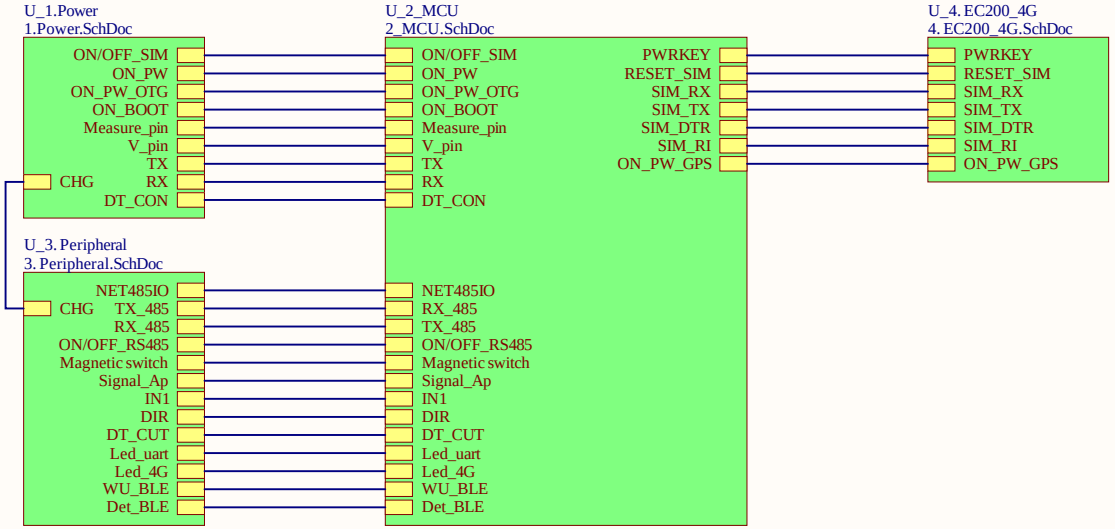
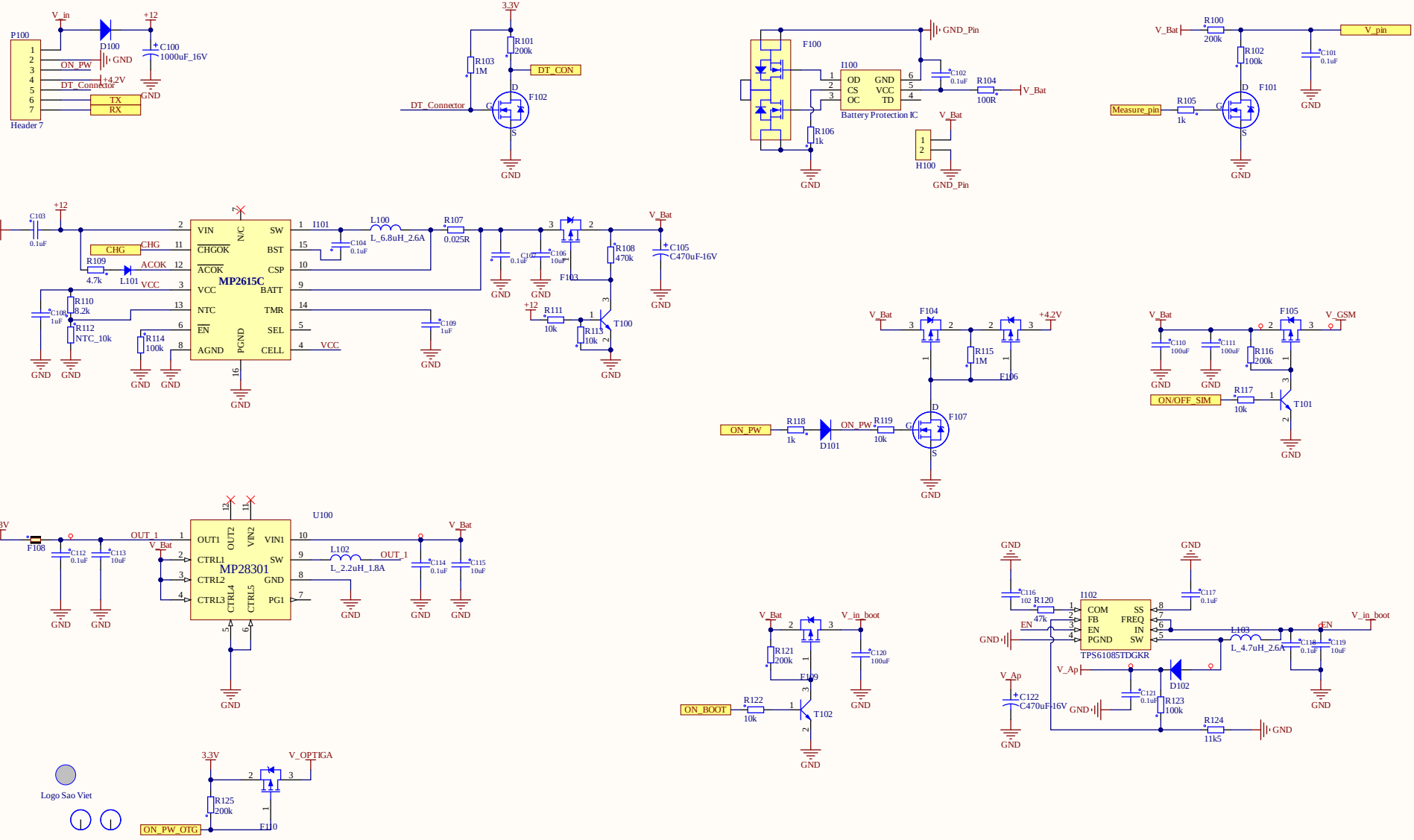
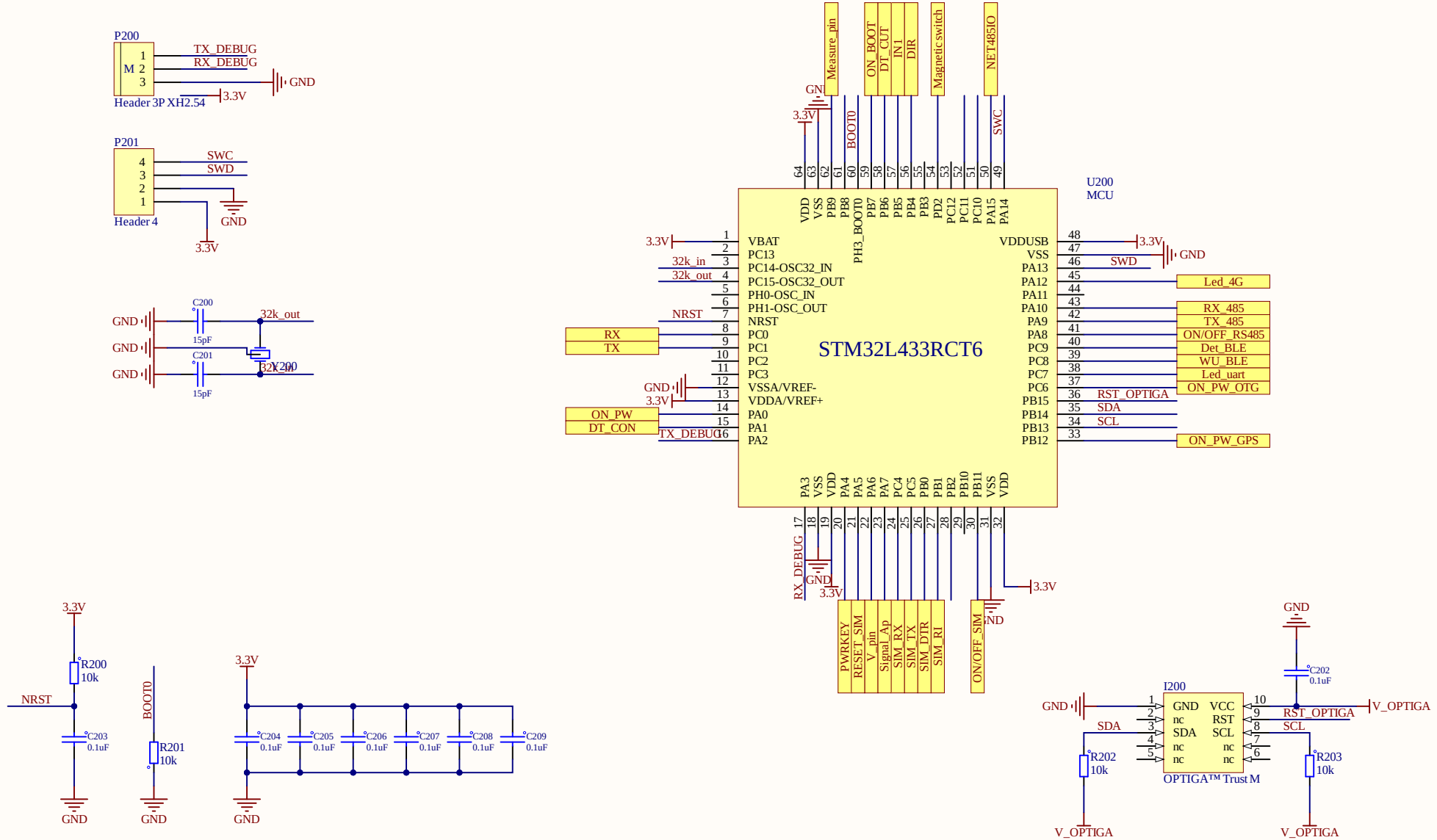
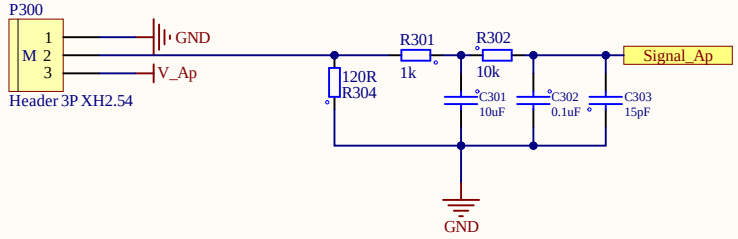
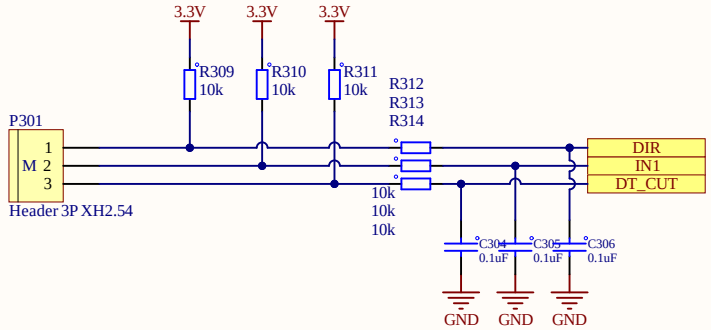


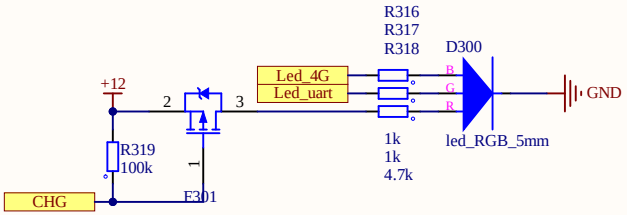
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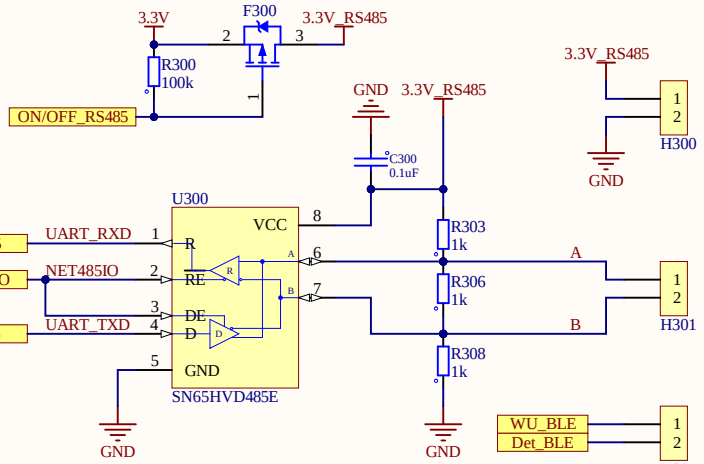

P100
Header 7
V_in
+12
D100
C100
1000uF_16V
ON_PW
+4.2V
DT_Connector
TX
RX
GND
3.3V
R101
200k
R103
1M
DT_CON
F102
F100
I100
Battery Protection IC
OD
GND
CS
VCC
OC
TD
R104
100R
R106
1k
H100
GND_Pin
V_Bat
R100
V_pin
R102
100k
C101
0.1uF
F101
Measure_pin
R105
1k
C103
R109
4.7k
L101
CHG
ACOK
VCC
MP2615C
VIN
CHGOK
ACOK
VCC
NTC
EN
AGND
N/C
SW
BST
CSP
BATT
TMR
SEL
CELL
PGND
I101
L100
L_6.8uH_2.6A
R107
0.025R
C104
C105
C470uF-16V
F103
R108
470k
R110
8.2k
R112
NTC_10k
R114
100k
C108
C109
R111
10k
R113
T100
F104
F106
R115
1M
F107
R118
D101
R119
C110
C111
C116
C117
F105
V_GSM
R116
200k
R117
ON/OFF_SIM
T101
U100
MP28301
OUT1
OUT2
CTRL1
CTRL2
CTRL3
CTRL4
CTRL5
VIN2
VIN1
SW
GND
PG1
F108
C112
C113
OUT_1
L102
L_2.2uH_1.8A
C114
C115
R121
200k
F109
C120
100uF
ON_BOOT
R122
T102
V_in_boot
C122
C470uF-16V
I102
TPS61085TDGKR
COM
FB
EN
PGND
SS
FREQ
IN
SW
R120
47k
L103
L_4.7uH_2.6A
C118
C119
D102
R123
100k
R124
11k5
V_Ap
Logo Sao Viet
R125
200k
F110
ON_PW_OTG
V_OPTIGA

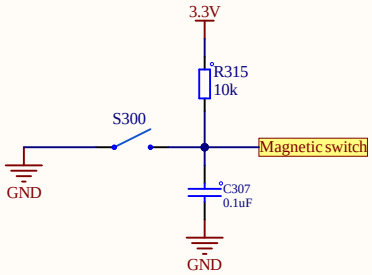

P200
M
1
2
3
TX_DEBUG
RX_DEBUG
GND
3.3V
Header 3P XH2.54
P201
4
3
2
1
SWC
SWD
GND
3.3V
Header 4
C200
15pF
C201
15pF
32k_out
32k_in
GND
U200
MCU
STM32L433RCT6
Measure_pin
ON_BOOT
DT_CUT
IN1
DIR
Magnetic switch
NET485IO
BOOT0
SWC
SWD
VBAT
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
NRST
PC0
PC1
PC2
PC3
VSSA/VREF-
VDDA/VREF+
PA0
PA1
PA2
PH3_BOOT0
VDDUSB
VSS
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PB15
PB14
PB13
PB12
RX
TX
ON_PW
DT_CON
Led_4G
RX_485
TX_485
ON/OFF_RS485
Det_BLE
WU_BLE
Led_uart
ON_PW_OTG
RST_OPTIGA
SDA
SCL
ON_PW_GPS
PWRKEY
RESET_SIM
V_pin
SignalAp
SIM_RX
SIM_TX
SIM_DTR
SIM_RI
ON/OFF_SIM
R200
10k
NRST
C203
0.1uF
R201
10k
BOOT0
C204
0.1uF
C205
0.1uF
C206
0.1uF
C207
0.1uF
C208
0.1uF
C209
0.1uF
I200
OPTIGA™ Trust M
GND
nc
SDA
SCL
nc
nc
VCC
RST
SCL
nc
nc
C202
0.1uF
R202
10k
R203
10k
V_OPTIGA
RST_OPTIGA



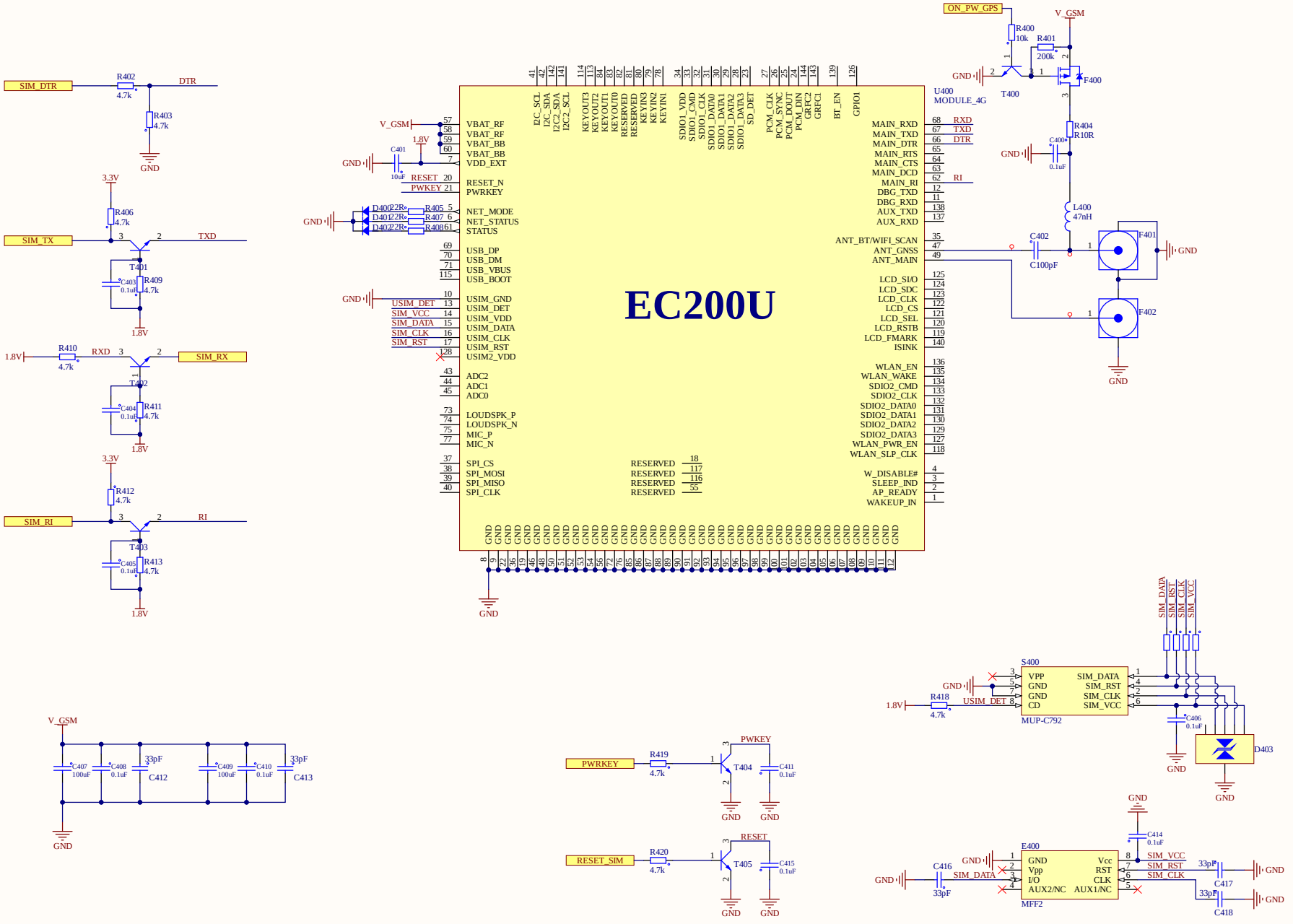


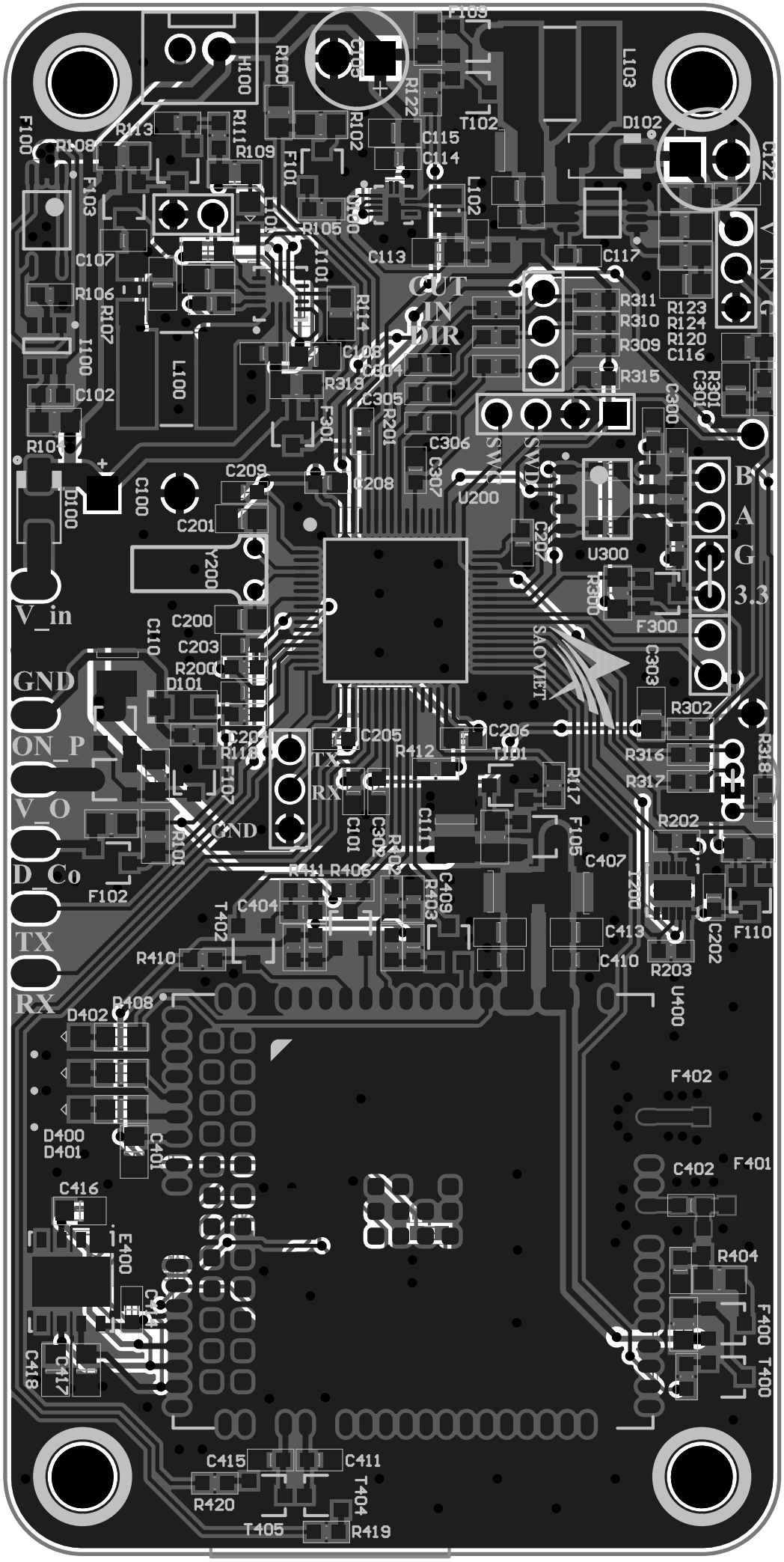






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EC200U
U400
MODULE_4G
VBAT_RF
VBAT_BB
VDD_EXT
RESET_N
PWRKEY
NET_MODE
NET_STATUS
STATUS
USB_DP
USB_DM
USB_VBUS
USB_BOOT
USIM_GND
USIM_DET
USIM_VDD
USIM_DATA
USIM_CLK
USIM_RST
USIM2_VDD
ADC2
ADC1
ADC0
LOUDSPK_P
LOUDSPK_N
MIC_P
MIC_N
SPI_CS
SPI_MOSI
SPI_MISO
SPI_CLK
RESERVED
I2C_SCL
I2C_SDA
I2C2_SDA
I2C2_SCL
KEYOUT3
KEYOUT2
KEYOUT1
KEYOUT0
KEYIN3
KEYIN2
KEYIN1
SDIO1_VDD
SDIO1_CMD
SDIO1_CLK
SDIO1_DATA0
SDIO1_DATA1
SDIO1_DATA2
SDIO1_DATA3
SD_DET
PCM_CLK
PCM_SYNC
PCM_DOUT
PCM_DIN
GRFC2
GRFC1
BT_EN
GPIO1
MAIN_RXD
MAIN_TXD
MAIN_DTR
MAIN_RTS
MAIN_CTS
MAIN_DCD
MAIN_RI
DBG_TXD
DBG_RXD
AUX_TXD
AUX_RXD
ANT_BT/WIFI_SCAN
ANT_GNSS
ANT_MAIN
LCD_SI/O
LCD_SDC
LCD_CLK
LCD_CS
LCD_SEL
LCD_RSTB
LCD_FMARK
ISINK
WLAN_EN
WLAN_WAKE
SDIO2_CMD
SDIO2_CLK
SDIO2_DATA0
SDIO2_DATA1
SDIO2_DATA2
SDIO2_DATA3
WLAN_PWR_EN
WLAN_SLP_CLK
W_DISABLE#
SLEEP_IND
AP_READY
WAKEUP_IN
GND
V_GSM
1.8V
RESET
PWKEY
RXD
TXD
DTR
RI
SIM_DTR
SIM_TX
SIM_RX
SIM_RI
ON_PW_GPS
T400
F400
L400
47nH
F401
F402
C100pF
R10R
T401
T402
T403
T404
T405
S400
MUP-C792
VPP
CD
SIM_DATA
SIM_RST
SIM_CLK
SIM_VCC
USIM_DET
D403
E400
MFF2
Vpp
I/O
AUX2/NC
AUX1/NC
Vcc
RST
CLK
C412
C413
C417
C418
33pF
100uF
0.1uF
10uF
4.7k
10k
200k
R10R
47nH


V_in
GND
ON_P
V_O
D_Co
TX
RX
V IN G
B A G 3.3
SAO VIET
SWD
OUT
IN
DIR
TX
RX
GND

Comment	Description	Designator	Footprint	LibRef	Quantity	Manufacturer Part Number
6R	6R	R305, R307, R414, R415, R416, R417	R0603	R4.7k_0805	6	
0.025R	Res 25 mrOhm 1% 0.5W 2205	Rcurr	R1206	R025t_1206	1	PE1206R0R47R5ZS TL
0.1uF	Cap SMT Cer .1UF 50V 10% 0603 4M TR	C100, C102, C103, C104, C107, C112, C114, C117, C118, C121, C202, C203, C204, C205, C206, C207, C208, C209, C300, C304, C305, C306, C307, C400, C403, C404, C405, C406, C408, C410, C411, C414, C415,	C0603	C0.1uF-0805	34	GMC10X7R104K50 NT
5k	RES SMD 1K OHM 5% 1/8W 0805	R206, R208, R118, R203, R203, R306, R308, R316, R317	R0603	R5k_0805	9	CR0805-8W-1022T
10M	SMD 470K OHM 1% 1/8W 0805	R103, R115	R0805	R200k_0805	2	
1uF	0.10µF 50V Ceramic Capacitor X7R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm)	C106, C109	C0805	C1uF_0805	2	TMK212B7105KG-T
4.7k	4.7k Ohm ±1% 0.125W, 1/8W Surface Mount Resistor Thick Film ±100ppm/°C 0805	R109	R0805	R4.7k_0805	1	RC0805FR-074K7L
4.7k	RES SMD 1K OHM 5% 1/8W 0805, 4.7K Ohm ±1% 0.125W, 1/8W Surface Mount Resistor Thick Film ±100ppm/°C 0805	R218, R402, R403, R406, R409, R410, R411, R412, R413, R418, R419, R420	R0603	R5k_0805, R4.7k_0805	12	CR0805-8W-1022T, RC0805FR-074K7L
8.2k	RES SMD 8.2K OHM 5% 1/8W 0805	R110	R0805	R8.2k_0805	1	RC0805JR-078K2L
10k	Res General Purpose Thick Film 0603 10K Ohm 1% 1/10W ±100ppm/°C Molded SMD Paper T/R, RES SMD 10K OHM 5% 1/16W 0603	R111, R113, R117, R119, R122, R200, R201, R203, R302, R309, R310, R311, R312, R313, R314, R315, R400	R0603	R10k_0603	18	RC0603FR-0710KL, RC1608J103CS	
10uF	10µF 10V Ceramic Capacitor X5R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm), 10µF 6.3V Ceramic Capacitor X5R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm)	C106, C113, C115, C116, C301, C401	C0805	C10uF_0805_10V, C10uF_0805	6	UMK212AB7106KG-T, AMK212AB7106KG HT
11k5	11.5k Ohm ±1% 0.125W, 1/8W Chip Resistor 0805	R134	R0805	R11.5k_0805	1	RC0805FR-0711K5L
15pF	15pF ±1% 50V Ceramic Capacitor C0G, NP0 0603	C205, C201, C303	C0805	C15pF-0805	3	08055C150FS00N T
22R		R405, R407, R408	R0805	R22R -0805	3	
32.768k	32.768kHz ±20ppm Crystal 12.5pF 50 kOhm -40°C ~ 85°C Surface Mount Cylindrical Can, Metal	Y200	TF-206	Crystal_EPSON_32.76 8_9pF-20ppm	1	EC3 - 327-12.5-12K1-X-C
33pF	33pF ±1% 50V Ceramic Capacitor C0G, NP0 0805 (2012 Metric)	C412, C413, C416, C417, C418	C0805	C33pF-0805	5	CC0805FRNPO9B X330
47k	47kOhms ±5% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thick Film	R135	R0805	R47k_0805	1	RC0805JR-074K7L
47nH	FIXED IND 47nH	L400	L0603	L4.7nJ -0603	1	
100k	RES SMD 100K OHM 5% 1/10W 0603	R132, R114, R123, R330, R319	R0805	R100k_0603	5	[NoParam], RC0603JR-07100KL
100R		R134	R0805	R100R	1	ERJ-U08G101RV
100uF	Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% -15x1, 20% -15x1, X5R, 15% TC, 100uF, Surface Mount, 1206, 100µF 10V Ceramic Capacitor X5R 3216 1206 Metric) 0.126" L x 0.098" W (3.20mm x 2.50mm)	C110, C111, C120, C207, C409	C1210	C100uF_1210_10V	5	GMC31X5R107M6 03NT, UMJ325B7107MM-T
102	1000pF ±10% 100V Ceramic Capacitor X7R	C116	C0805	C1nF -0805	1	HMJ212KB7102KD HT
120R	120 Ohms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thick Film	R204	R0805	R120R	1	RC0805FR-07120RL
200k	SMD 200K OHM 1% 1/8W 0805	R130, R101, R116, R125, R125, R401	R0805	R200k_0805	6	RT0805FRD07200K
470k	SMD 470K OHM 1% 1/8W 0805	R139	R0805	R300k_0805	1	
700mA Step-Down	Ultra-Low 500mA IQ, Wide Input 2-5.5V, 700mA Step-Down Regulator Plus 300mA IQ, 100mA LDO	U100	MP28301	MP28301	1	MP28301GG-P
1000uF_16V	CAP ALUM 1000UF 20% 16V RADIAL	C100	C_10*17mm	C_330µF 16V	1	UVR1C102MPD1TD
Battery Management 2.1A, 1- or 2-Cell Li-Ion Battery Charg	Battery Charger Li-Ion(Li-Poly) 2.1A 1.95V to 18V 3A Pin 20% TYP	U101	MP2615CGQ-P	MP2615CGQ-P	1	MP2615CGQ-P
Battery Protection IC	One Cell Lithium-Ion/Polymer Battery Protection IC	U100	DW01	DW01	1	
C100pF	100pF ±5% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	C402	C0603	C100nF 0603 1%	1	GCM188C1H101FA16D
C470uF-16V	470µF 16V Aluminum Electrolytic Capacitors Radial, Can 2000 Hrs @ 85°C	C105, C122	CP_6*12mm	C470uF-16V	2	UVR1C471MPD1TD
Long bar 5a		S300	Magnetic switch	Magnetic switch	1	MH2CD83
Diode Schottky 20V 2A	Vr=20V If=500mA 175pF Pd=310mW	D101	SOD-L3EBK90-2N	Diode Schottky 20V 2A	1	B5S20LW-7-F
ESD-A5V1-5WS	TVS DIODE 3VWM SOT323S	D403	SOT23-6N -SMR5SC SOT363 TVS DIODE ESD PROTECTION SMR5SC SOT363 TVS DIODE ESD PROTECTION SMR5SC	TVS 3VC SOT-363	1	ESD-A5V1-5WS
Ferrite bead 0805	600R/TE BEAD 35 OHM 0805 1LN	L106	L0805	Ferrite bead 0805	1	CB21F150N0
H7	Hole_3mm	*H20, *H2L, *H22, *H3	Hole_3mm	hole_4mm	4	
Header 2H_j	Header, 2-Pin, Right Angle	H100	header_2P_x52.54	Header 2H_j	1	
Header 2H_j	Header, 2-Pin, Right Angle	H300, H301, H302	HDR1X2	Header 2H_j	3	
Header 3P 1X52.54		FJ00, FJ00, FJ01	HDR1X3	Header 3P 1X2.54	3	
Header 4	Header, 4-Pin	FJ01	HDR1X4	Header 4	1	
Header 7	Header, 7-Pin	FJ05	Connector_7	Header 7	1	
IRLML6246TRPBF	MOSFET N-CH 20V 6A SOT-23	F101, F102, F107	SOT23A	IRLML6246TRPBF	3	IRLML6246TRPBF
IRML6401PbF	MOSFET P-CH 12V 4.3A SOT-23	F103, F104, F105, F106, F108, F110, F300, F301, F400	SOT23A	IRML6401PbF	9	IRLML6401TRPBF
Khe Antien Receptacle U.FL		R401, R402	Khe Antien Receptacle U.FL	Khe Antien Receptacle U.FL	2	
L_2.2uH_1.8A	Inductor Power Thin Film 2.2uH 20% 1MHz Metal 1.8A 0.13Ohm DCR 0806 T/R	L102	L2016	L_2.2uH	1	CSCT201610U-H2R2 MNE
L_4.7uH_2.6A	4.7uH ±20% 0.0403Ω 39MHz 6.0x6.0mm, NR6028	L103	Inductor-6x6mm	L_4.7uH	1	NR6028T4R7M
L_6.8uH_2.6A	SMD Power Inductors 6028 6.8UH 2	L100	Inductor-6x6mm	L_6.7uH	1	AMPD6030G6R8 MT
LED_BLUE	Blue 470nm LED Indication - Discrete 0.85V 0805	D400, D401, D402	LED0805_Blue	LED_0805_bl	3	
led_RGB_5mm		D200	LED_RGB_pin-5	LED_RGB	1	
Logo		Logo Sao viet100	Logo Sao viet	Logo	1	
MCU	Ultra-low-power with FPU ARM Cortex-M4 MCU 80 MHz with 256 Kbytes Flash, LCD, USB	U200	LQFP64	STM32L433RCT6	1	STM32L433RCT6
MP2		K400	SSIM_MP2	SSIM	2	
MMBT3904LT1G	TRANS NPN 40V 0.2A SOT23	T400, T101, T102, T403, T402, T402, T403, T404, T405	SOT23A	NPN- MMBT3904LT1G	9	MMBT3904LT1G
MOD1p5_4G		Ku400	MC20SU	MC20SU	1	
MOSFET N-2CH		K100	FDW256A	FDW256A	1	
MUP_C702		S402	Card_sim_micro	Micro_sim	1	
NTC_10k	10k Thermistor, 10Kohm, Radial Leaded Rohs Compliant: Yes	R112	HDR1X2	R5k_0805	1	NXRT15XH103FA1 B030
OPTIGA™ Trust M		U202	OPTIGA™ Trust M	OPTIGA™ Trust M	1	
R10R	RES SMD 10 OHM 10% 1/8W 0805	R404	R0805	R10R -0805	2	CR0805JR-0710RL
SN65HVD485E	IC TXRX RS485/RS422 8-SOIC	U300	Sop-8	MAX485 SN65HVD72DR	1	SN65HVD485E
SS34	40V PURP 1AV 1A SMA	D100, D102	DO-214AC	Diode_SCHOKY 1AV 1A SMA	2	SS34-HF
TP1610087DG08		U102	TP1610087DG08	TP1610087DG08	1	TP1610087DG08
Value		L101	LED0805_Blue	LED_3mm	1	